Robust Local Likelihood Estimation

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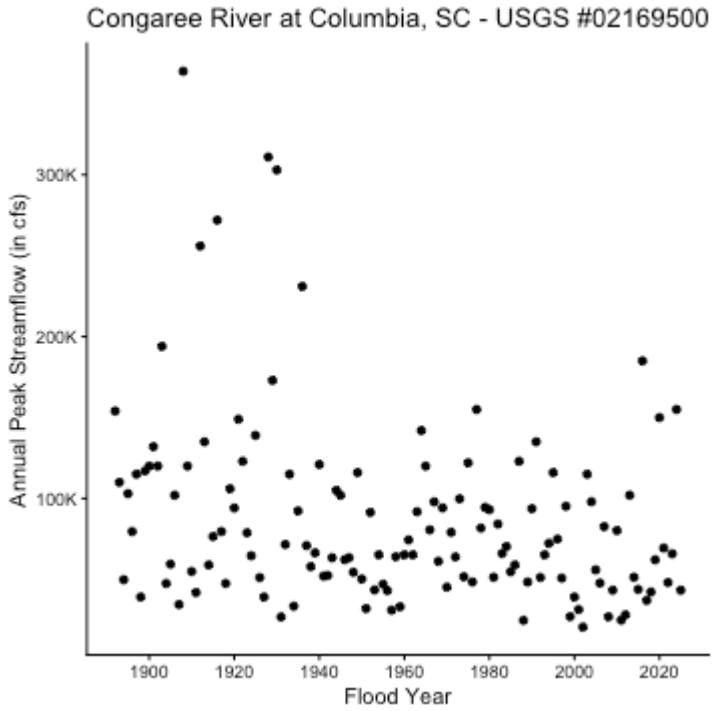
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Robust Local Likelihood Estimation and Flood Frequency Analysis

Case Study: USGS #02169500

USGS #02169500 is the flood gage for the Congaree River at Columbia, South Carolina, USA. The annual peak streamflows, measured in cubic feet per second, are displayed for the flood years 1892 to 2025.



Flood Frequency Analysis

Often we are interested in a specific quantile from a distribution of the annual peak streamflows.

$$P(X > Q_T) = \int_{Q_T}^{\infty} f(x)\, dx = \frac{1}{T}$$

For flood frequency analysis in the USA, Q_{100} is of particular interest. This is the **1% chance flood (100-year flood).**

- Bulletin 17C suggests using a Pearson Type III distribution for the $\log_{10}$-transformed annual peak streamflows

For flood frequency analysis in the UK, Q_{MED}, the **median** of the annual peak streamflows, is widely used in various applications. Quantiles from the flood distribution are also of interest.

- FEH suggests using a generalized logistic distribution for the annual peak streamflows

In the hydrological literature, the generalized extreme value (GEV) distribution is commonly used.



GEV Distribution

The pdf for the generalized extreme value distribution is

$$f(x) = \frac{1}{\sigma}\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi} \exp\left\{-\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]\right\}$$

- μ is the **location** parameter $(\mu \in \mathbb{R})$
- σ is the **scale** parameter $(\sigma > 0)$
- ξ is the **shape** parameter $(\xi \in \mathbb{R})$

The **support** for this distribution depends on ξ:

- if $\xi > 0$ then $x \in \left[\mu - \frac{\sigma}{\xi}, +\infty\right)$
- if $\xi = 0$ then $x \in (-\infty, \infty)$
- if $\xi < 0$ then $x \in \left(-\infty, \mu - \frac{\sigma}{\xi}\right]$



Robust Local Likelihood

A big assumption in flood frequency analysis is that the annual peak streamflows represent a **stationary** process.

- Due to climate change, some researchers bluntly say that "stationarity is dead" (Milly et al. 2008)

Robust Local Likelihood:

Suppose annual peak streamflows, y_i, $i = 1, \ldots, n$, are observed at time points $t_i = i/n$, $t \in (0, 1)$

Step 1: Fit the local likelihood function at time point t.

$$L\left(\boldsymbol{\Theta}_t\right) = \sum_{i=\max(1,[n(t-h)])}^{\min(n,[n(t+h)])} w_{i,t} \log f\left(y_i; \mu_{it}, \sigma_t, \xi_t\right)$$

- $f(\cdot;\cdot)$ is a GEV density function with parameters $\boldsymbol{\Theta}_t = (\beta_{0,t}, \beta_{1,t}, \gamma_t, \xi_t)$ where a linear model for the location parameters is $\mu_{it} = \beta_{0,t} + \beta_{1,t}(i - nt)$ and the scale parameter is $\sigma_t = \exp[\gamma_t]$

Robust Local Likelihood

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- For local weights $\{w_{i,t}\}$, the Epanechnikov kernel,

$$w_{i,t} = \left(1 - \frac{t_i - t}{h}\right)^2$$

  is used, with h controlling the window width

Step 2: Compute residuals $e_{i,t} = y_i - \hat{\mu}_{i,t}(x)$ using the estimated mean of the distribution.

$$\hat{\mu}_{i,t}(x) = \hat{\mu}_{it} + \frac{\hat{\sigma}_t}{\hat{\xi}_t}\left(\Gamma\left(1 - \hat{\xi}_t\right) - 1\right)$$



Robust Local Likelihood

Step 3: Once the local likelihood mean and residuals have been fit, set $s = \text{med}|e_{i,t}|$ and define robustness weights

$$v_{i,t} = B\left(\frac{e_{i,t}}{6s}\right)$$

where $B(u) = \left(1 - u^2\right)^2, 0 \leq u \leq 1.$

Step 4: Re-estimate $\boldsymbol{\Theta}_i$ using updated weights $w^*_{i,t} = w_{i,t}v_{it}$.

- The process can be iterated until estimates converge

The Hope:

- The expectation is that a substantively different estimate of $\boldsymbol{\Theta}_i$ may be derived than obtained by a single local likelihod pass.
- How? The largest floods in the series could be "down-weighted", which could be considered as arising from a separate cause than other peak flows in the series.

Using a bandwidth of $h = 0.325$ and the `RLLgev.R` function on the Congaree annual peak streamflows:

```
y ← congaree$Peak_Flow
n ← length(y)
t ← 1:n/n

congareeRLL ← RLLgev(t,y,0.325,40000)

congaree.results ← as.data.frame(congareeRLL$parm.est) %>%
  mutate(Year=congaree$Year,Peak=congaree$Peak_Flow)

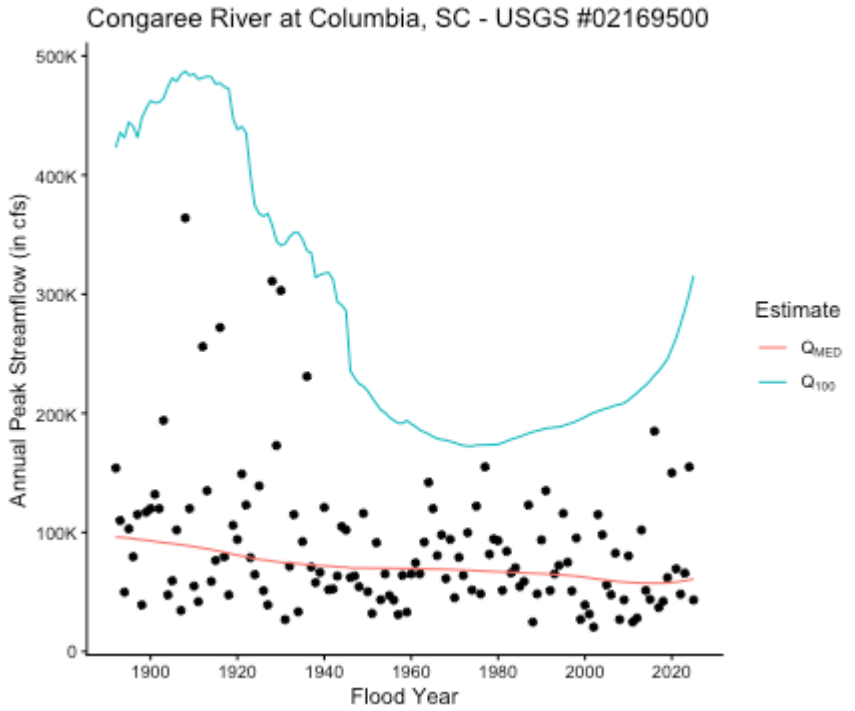
# Reshaped data set for plotting purposes
congaree.long ← congaree.results %>%
  select(Year,Median,Q100) %>%
  pivot_longer(-Year,names_to="Estimate",values_to="cfs")

ggplot(congaree.results,aes(x=Year,y=Peak))+geom_point()+xlab("Flood Year")+
  ylab("Annual Peak Streamflow (in cfs)")+theme_classic()+
  geom_line(data=congaree.long,aes(x=Year,y=cfs,color=Estimate))+
  scale_y_continuous(breaks=c(seq(0,500000,100000)),
                     labels=c("0","100K","200K","300K","400K","500K"))+
  scale_x_continuous(breaks=c(seq(1900,2020,20)))+
  ggtitle("Congaree River at Columbia, SC - USGS #02169500")+
  scale_color_discrete(labels=c(expression(Q[MED]),expression(Q[100])))
```



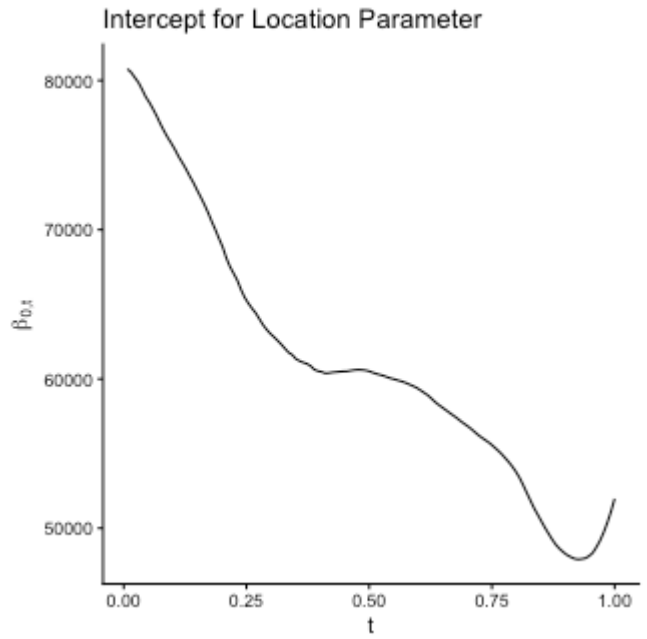
Case Study: USGS #02169500

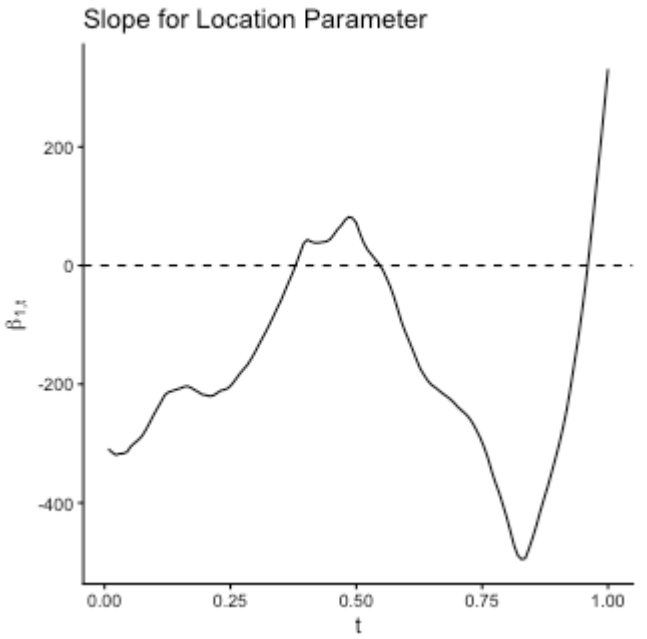
USGS #02169500 is the flood gage for the Congaree River at Columbia, South Carolina, USA. The annual peak streamflows, measured in cubic feet per second, are displayed for the flood years 1892 to 2025.



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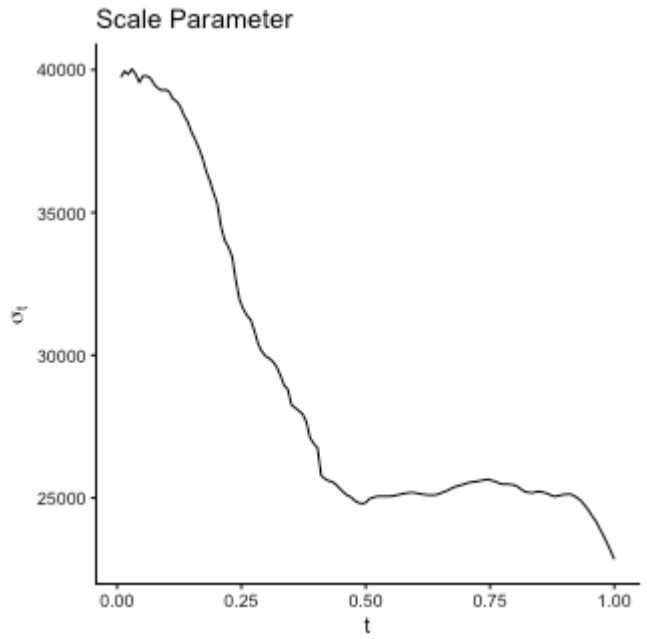
We can also investigate how the parameter estimates change over time t. Recall that $\mu_{it} = \beta_{0,t} + \beta_{1,t}(i - nt)$.

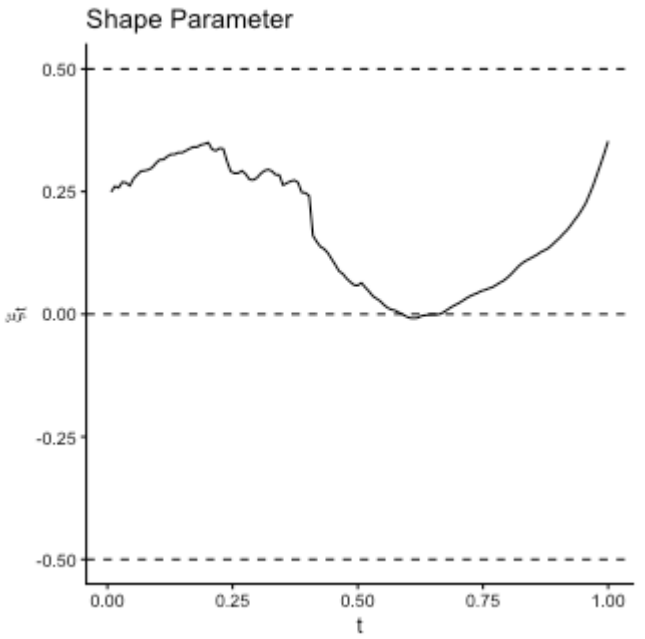




Case Study: USGS #02169500

Recall that $\sigma_t = \exp[\gamma_t]$. Past flood frequency studies using the GEV expect the shape parameter to be less than 0.5 in magnitude.





Case Study: USGS #02169500

We can also plot the robust local likelihood weights associated with each flood year.

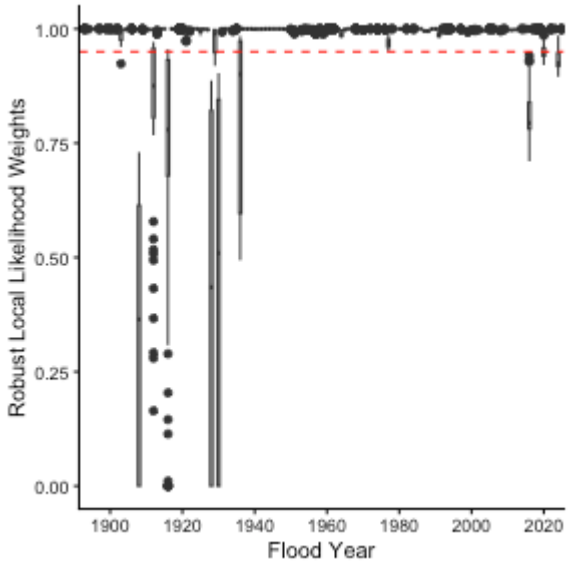
In the `$wts` produced by the `RLLgev.R` function:

- each **column** represents a **flood year's robust weights**
- each **row** represents the **weights used in a flood year's smoothing window**

```
# Plot the robust local likelihood weights
congaree.wts ← cbind(congareeRLL$wts,row.num=1:134)
congaree.wts.long ← as.data.frame(congaree.wts) %>%
  pivot_longer(-row.num,values_to="Weight") %>%
  mutate(Year=rep(1892:2025,134)) %>% select(-name,-row.num) %>%
  filter(!is.na(Weight))
ggplot(congaree.wts.long,aes(x=factor(Year),y=Weight))+geom_boxplot()+
  theme_classic()+ylab("Robust Local Likelihood Weights")+
  scale_x_discrete(breaks=c(seq(1900,2020,20)))+
  geom_hline(yintercept = 0.95,color="red",lty=2)+xlab("Flood Year")
```

Case Study: USGS #02169500

An arbritrarily chosen cutoff to identify "down-weighted" peak streamflows in the flood series is a median robust weight less than 0.95.



Nine flood years had their peak streamflows down-weighted:

- 1908, 1912, 1916, 1928, 1930, 1936, 2016, 2020, 2024

Standard Errors

The previous slides discussed how to find estimates for Q_{100} and Q_{MED}. Often standard errors of these point estimates are of interest.

Sandwich Estimator & The Delta Method (for a time point t)

- Find the Jacobian for the robust local likelihood estimator

$$J_i(t) = -\sum_{i=1}^{n} w_{it} \frac{\partial^2 l\left(\boldsymbol{\Theta}_i; y_i\right)}{\partial \boldsymbol{\Theta}_i \partial \boldsymbol{\Theta}_i^T}, \quad i = 1, \ldots, n,$$

  where $l\left(\boldsymbol{\Theta}_i; y_i\right)$ is the log-likelihood contribution at observation y_i and w_{it} represents the final robust set of weights

- The sandwich estimator of the variance-covariance matrix for $\hat{\boldsymbol{\Theta}}_i$ is

$$V_i(t) = J_i^{-1}(t) \left\{ \sum_{i=1}^{n} w_{it}^2 \frac{\partial l\left(\boldsymbol{\Theta}_i; y_i\right)}{\partial \boldsymbol{\Theta}_i} \frac{\partial l\left(\boldsymbol{\Theta}_i; y_i\right)}{\partial \boldsymbol{\Theta}_i^T} \right\} J_i^{-1}(t), \quad i = 1, \ldots, n$$

Standard Errors

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Sandwich Estimator & The Delta Method (for a time point t)

- The quantile F_q at t^* can be written as

$$F_q = \beta_{0,t} + \beta_{1,t}\,(t^* - t) + \frac{\exp[\gamma_{0,t}]}{\xi_t}\left[(-\log q)^{-\xi_t} - 1\right],$$

  where the quantile is estimated at $t^* = t$ (to simplify the form)

- The gradient of the quantile is

$$\nabla_{F_q} = \left(\frac{\partial F_q}{\partial \beta_{0,t}}, \frac{\partial F_q}{\partial \beta_{1,t}}, \frac{\partial F_q}{\partial \gamma_t}, \frac{\partial F_q}{\partial \xi_t}\right)^T$$

- The standard error of the estimated quantile is $\sqrt{\nabla_{F_q}^T V_i(t) \nabla F_q}$.



Future Work

Robust local likelihood methods can be potential used to model multi-source annual peak flows that more fully accounts for the risk of extreme flood events.

Consider a marginal pooled density function for flood frequencies:

$$f\left(y_i|\mathbf{\Theta}_{1t_i}, \mathbf{\Theta}_2\right) = \tau f_1\left(y_i; \mathbf{\Theta}_{1t_i}\right) + (1-\tau) f_2\left(y_i; \mathbf{\Theta}_2\right)$$

Here $f_1(\cdot)$ is a local linear GEV model while $f_2(\cdot)$ is a global GEV model with parameter vector $\mathbf{\Theta}_2 = (\mu^*, \sigma^*, \xi^*)^T$ representing the distribution of the most extreme events in the annual peak flow series.

The pooling proportion τ can be modeled as:

$$\tau = \sum_{i=1}^{n} E\left(z_i\right),$$

where z_i is an indicator for pooling events modeled by $f_1(\cdot)$.

Future Work

How do we find the pooling component τ?

Option #1: Robust Local Likelihood

- The down-weighted floods can be assigned to the second component and then $f_2(\cdot)$ is estimated with a GEV distribution
- The rest of the observations are in the first component and $f_1(\cdot)$ is estimated via robust local likelihood estimation

Option #2: Leave the Pooling Assignment Unspecified

- For each observation i, define the indicator vector $\mathbf{z}_i = (z_{i1}, z_{i2})^T$, with 1 in the position corresponding to the appropriate pooling component and 0's elsewhere
- For group j, the density function for y_i can be written as

$$f\left(y_i|\mathbf{z}_i = \mathbf{e}_j\right) = f_j\left(y_i|\mathbf{\Theta}_{tj}\right).$$

Future Work

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- The likelihood would actually need to specify a pooling distribution, $P(\mathbf{z})$, across all possible pooling sets $\mathbf{z} = (\mathbf{z}_1, \ldots, \mathbf{z}_n)$:

$$L\left(\eta\right) = \sum_{\mathbf{z}} P(\mathbf{z}) \left\{ \prod_{i=1}^{n} \left[\prod_{j=1}^{2} f_j\left(y_i|\mathbf{\Theta}_{t,j}\right)^{z_{ij}} \right] \right\}$$

- $P(\mathbf{z})$ could be modeled in a Bayesian framework like in certain model selection problems



References:

GitHub: https://github.com/philipayates/floods

England JF Jr, Cohn TA, Faber BA, Stedinger JR, Thomas WO Jr, Veilleux AG, Kiang JE, Mason RR Jr (2018), *Guidelines for determining flood flow frequency - Bulletin 17C (ver. 1.1, May 2019)*: U.S. Geological Survey Techniques and Methods, book 4, chap. B5, 148 p., https://doi.org/10.3133/tm4B5.

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Jóhannesson ÁV, Siegert S, Huser R, Bakka H, Hrafnkelsson B (2022) "Approximate Bayesian inference for analysis of spatiotemporal flood frequency data," *Annals of Applied Statistics*, 16(2): 905-935.

Milly, PCD, Betancourt J, Falkenmark M, Hirsch RM, Kundzewicz ZW, Lettenmaier DP, Stouffer RJ (2008) "Stationarity is dead: Whither water management?" *Science*, 319(1): 573-574.

UK Centre for Ecology & Hydrology, *Flood Estimation Handbook*, https://www.ceh.ac.uk/our-science/projects/flood-estimation-handbook.